

results. These sub-strings contain the names of the appliances you wish to control. A few conditional statements will check the availability of that string, that is, the presence of an appliance.

Suppose you wish to operate a TV. The code will detect its presence when you look at it. With the help of an if condition, time will be counted while you are looking at it. After five seconds the TV will switch on or switch off.

Raspberry Pi Pico coding

We need to create code for Raspberry Pi Pico so that it may connect to the main Raspberry Pi and control the device wirelessly we are looking at.

We will use micropython for Raspberry Pi Pico. We use USART to communicate with the Bluetooth HC 05 module and then using if condition check the commands to toggle the switch of connected appliances. Save the code as *main.py* to Raspberry Pi Pico.

Connection

Connect HC05, Pico, and the relay module as shown in Fig. 7. Connect the relay's COM pin to one wire of the electronic appliance and its normally-open pin to the live wire carrying AC voltage from the power supply. Connect the other wire of electric appliance to neutral wire of power supply. (**Warning:** Be care careful while working with AC power supply as wrong connection might cause electric shock and short circuit.)

You can add one more relay to 25-pin connector for phone and laptop charging.

Next, connect Raspberry Pi Pico to 5V DC adaptor and connect the Raspberry Pi 4 with a camera mounted on eyeglass with 5V battery and run the code *iotoneye.py* on Raspberry Pi 4. At EFY we included TV in the code for testing. When you look at TV for five seconds, it automatically turns on/off. Likewise, you can switch on/off any device. **EFY**

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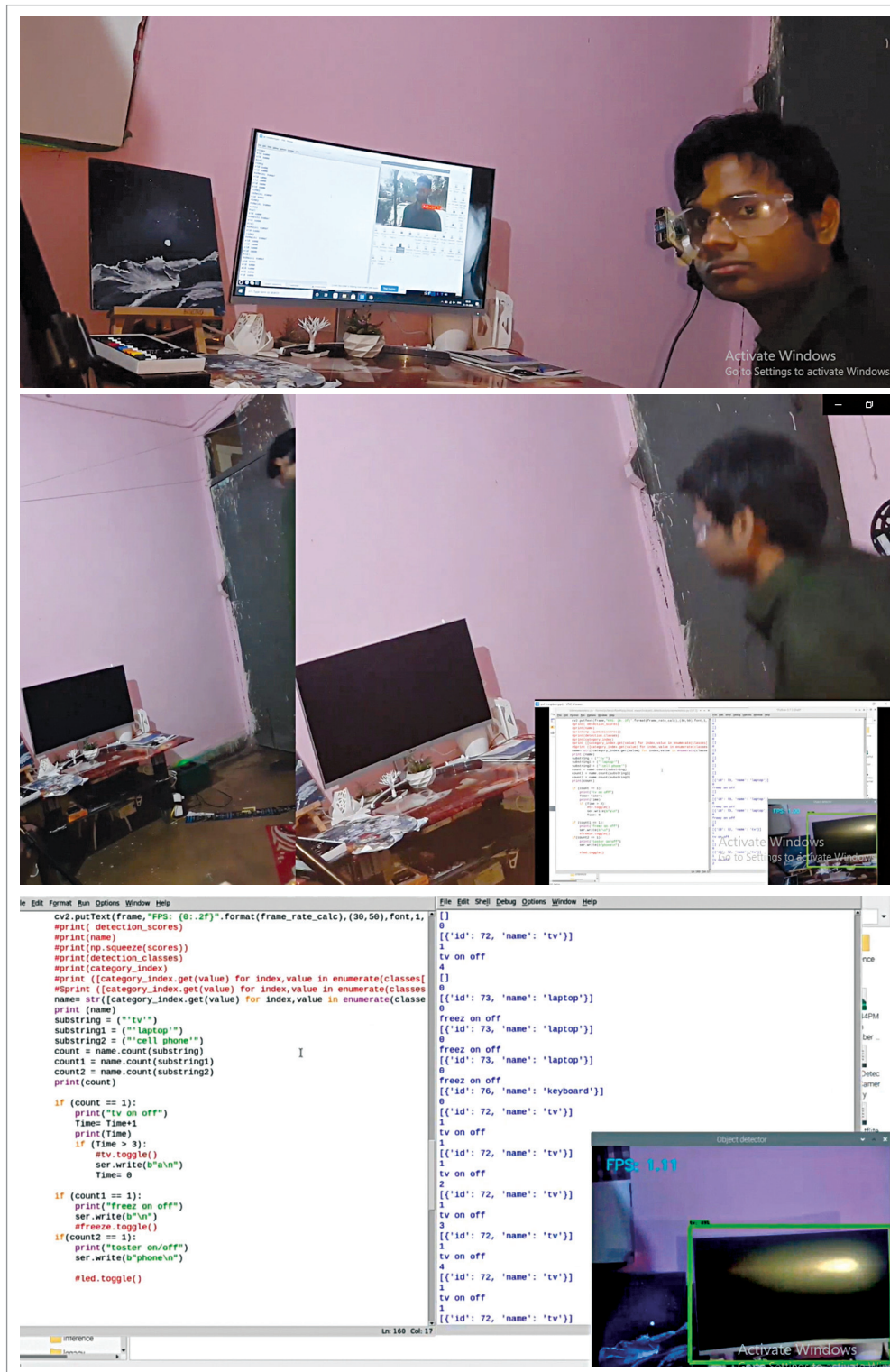


Fig. 8: Author testing the device